# GMapping-SLAM Mapping



## 1. Course Content

Basic: Run the sample program to perform SLAM mapping using the GMapping algorithm and save the grid map.

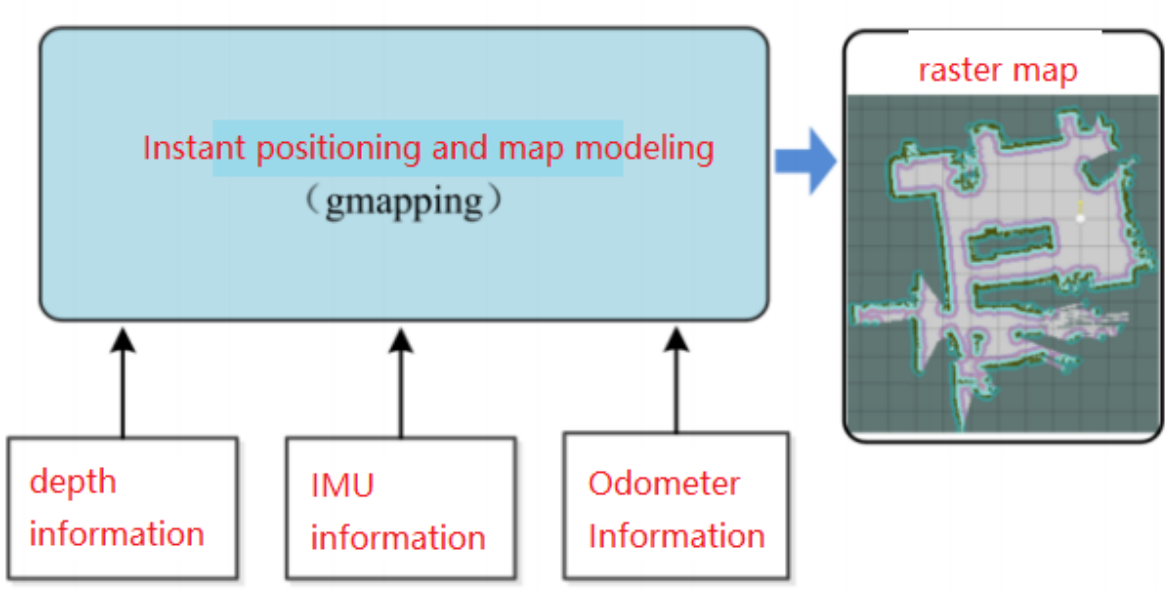
Advanced: Understand the configuration parameters of the GMapping algorithm.

## 2. GMapping Algorithm Introduction

- GMapping is only suitable for 2D laser scans with fewer than 1440 points per frame. If the number of points exceeds 1440, an issue like `[mapping-4] process has died` will occur.
- GMapping is a popular open-source SLAM algorithm based on the filter SLAM framework.
- GMapping is based on the RBPF (Rao-Blackwellized Particle Filter) algorithm, where the localization and mapping processes are separated. Localization is performed first, followed by mapping.
- GMapping makes two main improvements to the RBPF algorithm: an improved proposal distribution and selective resampling.

**Advantages:** GMapping can build indoor maps in real-time. It requires less computational power and offers higher accuracy for small-scale mapping.

**Disadvantages:** As the scene size increases, more particles are needed. Since each particle carries a map, both memory and computational requirements increase when building large maps. Therefore, it is not suitable for large-scale mapping. It also lacks loop closure detection, which can cause map misalignment when a loop is closed. Although increasing the number of particles can help close the map, it comes at the cost of increased computation and memory.



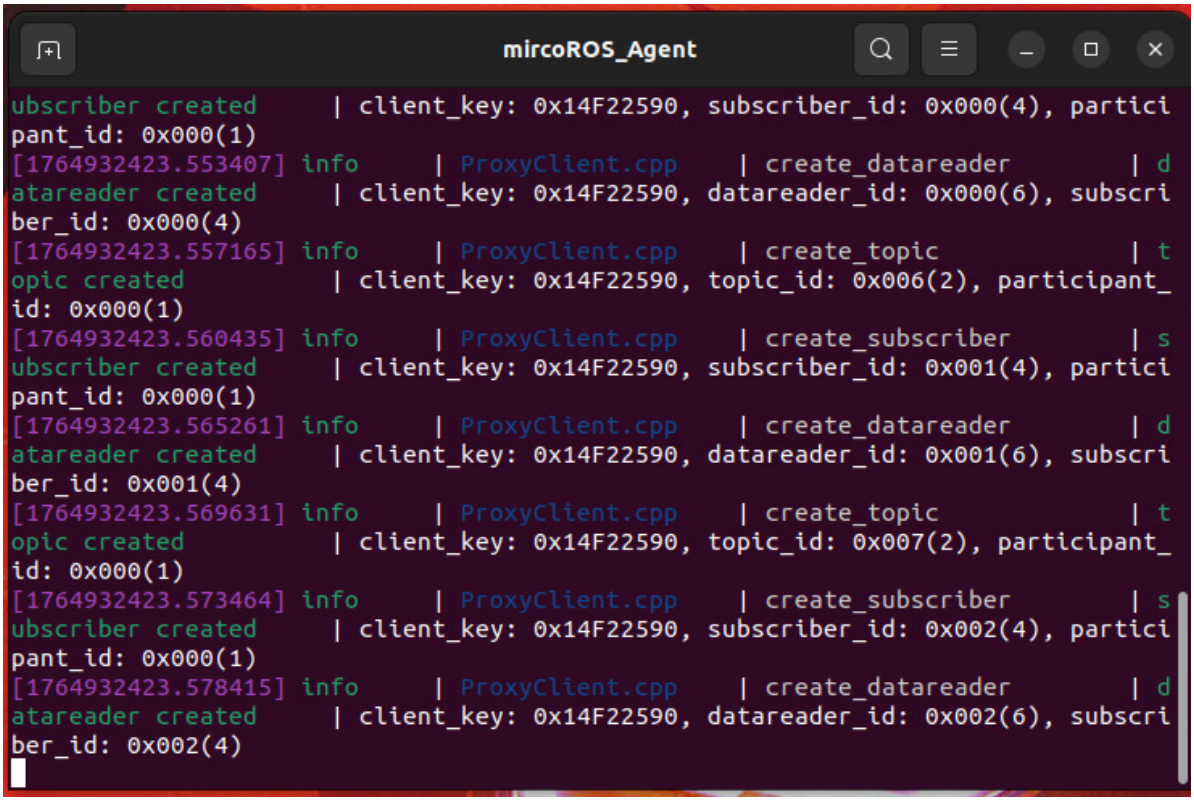

## 3. Start the Agent

**Note: If it is already running, you do not need to start it again.**

Enter the command in the vehicle's terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:




## 4. Run the Example

## 4.1 Mapping Process

- Note: When mapping, a slower speed (especially rotational speed) yields better results. If the speed is too high, the mapping quality will be poor.
- Launch the mapping command in the terminal:

**For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first,**

```
bringup_m3
```

**Open a terminal inside the container,**

```
exec_m3
```

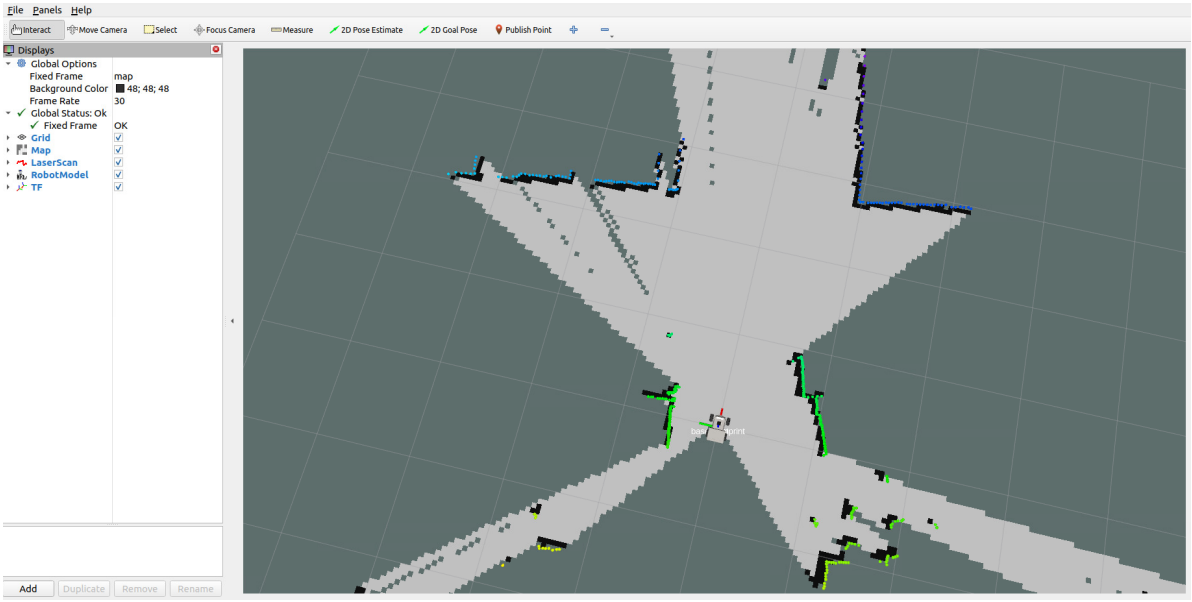
Enter the command,

(ORIN & PI5 boards)

```
ros2 launch m3_bringup gmapping.launch.py
```

(RDK board) Run in separate steps, with visualization launched on the accompanying virtual machine `Rosmaster-M3-Ubuntu22.04`,

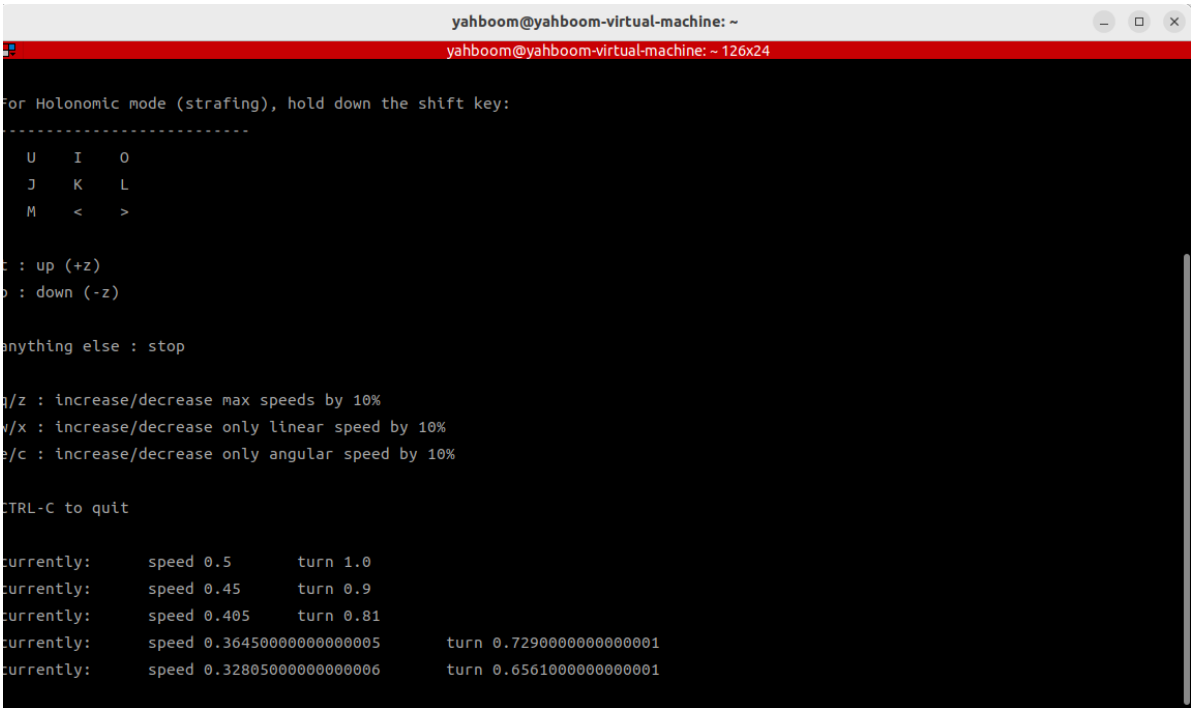
```
# On the virtual machine
rviz2 -d
/home/yahboom/yahboomM3_ws/user_ws/src/m3_bringup/rviz/gmapping_rviz.rviz
# On the host machine
ros2 launch m3_bringup gmapping.launch.py gui:=False
```



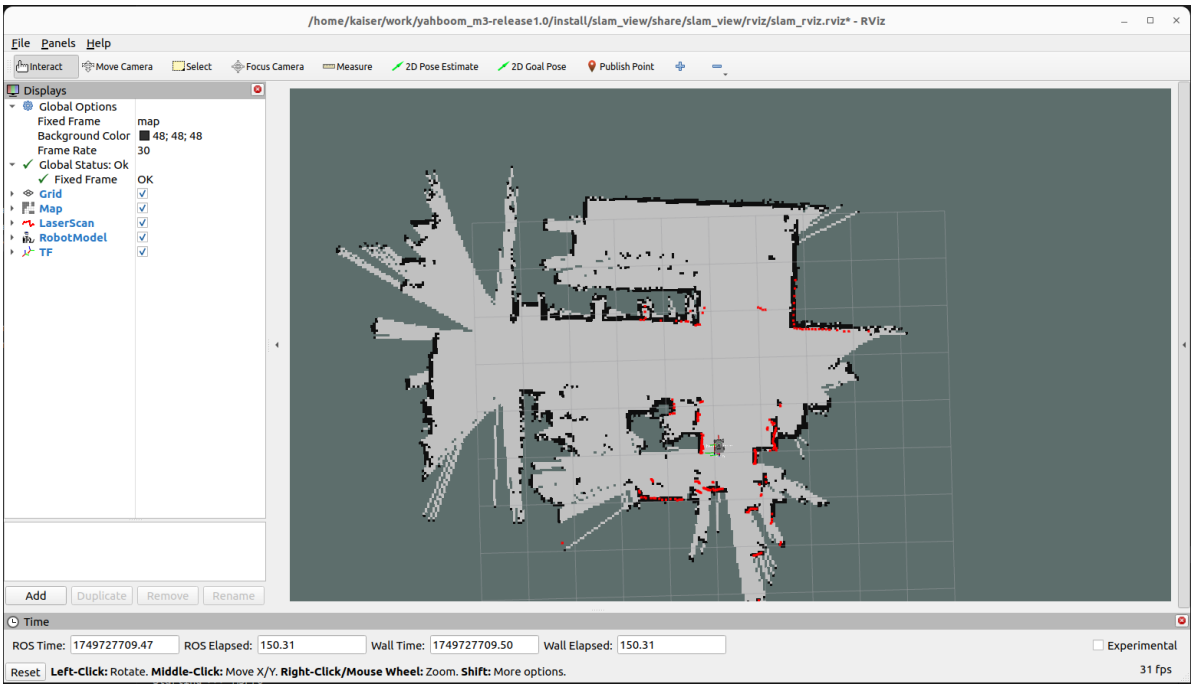

- To control the vehicle's movement, you can use either keyboard or joystick control. Here, we use keyboard control as an example:

```
keyboard_control
```

- Click on the terminal window with the mouse and press `z` to reduce the speed appropriately.




Press `I`, `<`, `J`, `L` to control the car to move forward, backward, turn left, and turn right, respectively, to complete the mapping.




## 4.2 Save the Map

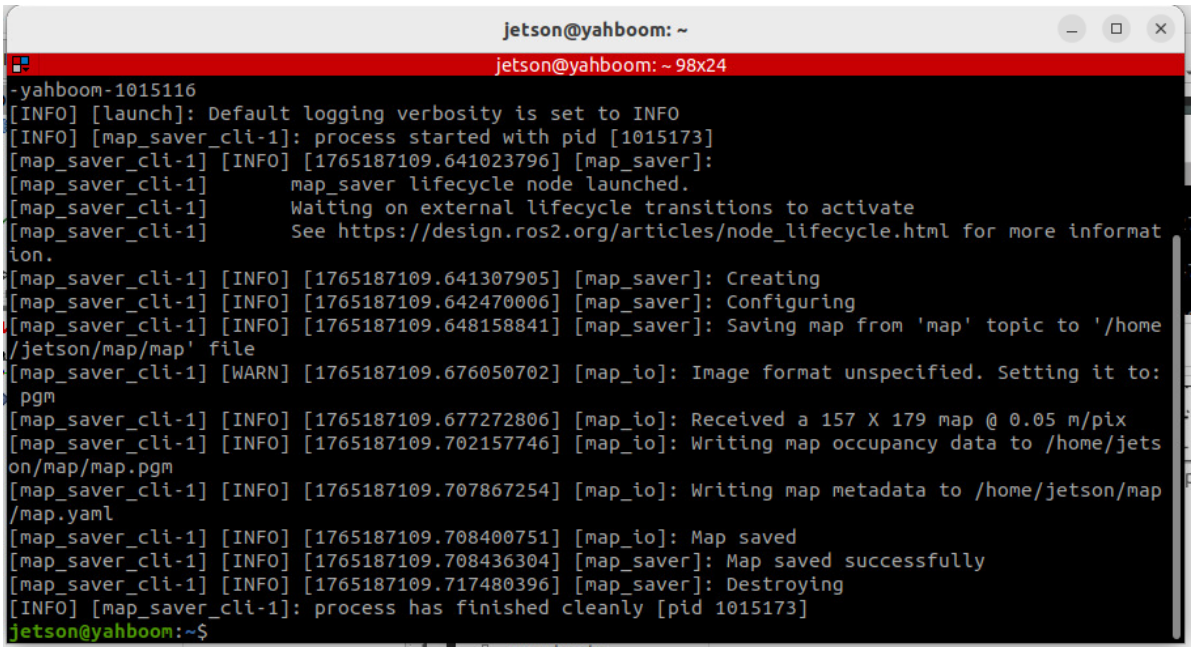
- Run the following command in the terminal to save the grid map:

```
ros2 launch m3_bringup save_map.launch.py
```

> [!TIP]
>
> **Optional Parameters**:
>
> `map_name`: Specifies the name of the saved map, with the default value being `map`.

> For example, to save a map named `test.yaml`, add the parameter: `ros2 launch m3_bringup save_map.launch.py map_name:=test`

The terminal prompt **Map saved successfully** indicates that the map was saved successfully. If it fails, try saving again.




The default map save path is as follows:

```
~/map/map.yaml
```

This includes a PGM image and a YAML file, `yahboom_map.yaml`.

```
image: yahboom_map.pgm
mode: trinary
resolution: 0.05
origin: [-10, -10, 0]
negate: 0
occupied_thresh: 0.65
free_thresh: 0.25
```

Parameter analysis:

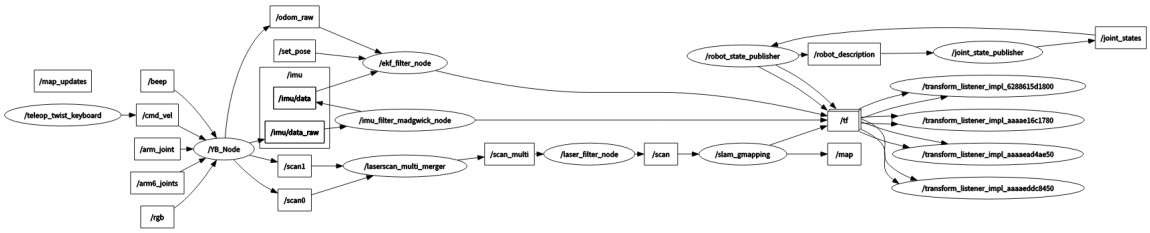
- `image`: The path to the map file, which can be an absolute or relative path.
- `mode`: This attribute can be `trinary`, `scale`, or `raw`, depending on the chosen mode. `trinary` is the default.
- `resolution`: The resolution of the map, in meters/pixel.
- `origin`: The 2D pose (x, y, yaw) of the bottom-left corner of the map. Here, `yaw` is a counter-clockwise rotation (yaw=0 means no rotation). Many parts of the system currently ignore the yaw value.
- `negate`: Whether to invert the meaning of white/black and free/occupied (the interpretation of thresholds is not affected).
- `occupied_thresh`: Pixels with an occupancy probability greater than this threshold are considered fully occupied.
- `free_thresh`: Pixels with an occupancy probability less than this threshold are considered fully free.

# 5. Node Analysis

## 5.1 Display Node Computation Graph

Run in the virtual machine terminal:

```
ros2 run rqt_graph rqt_graph
```

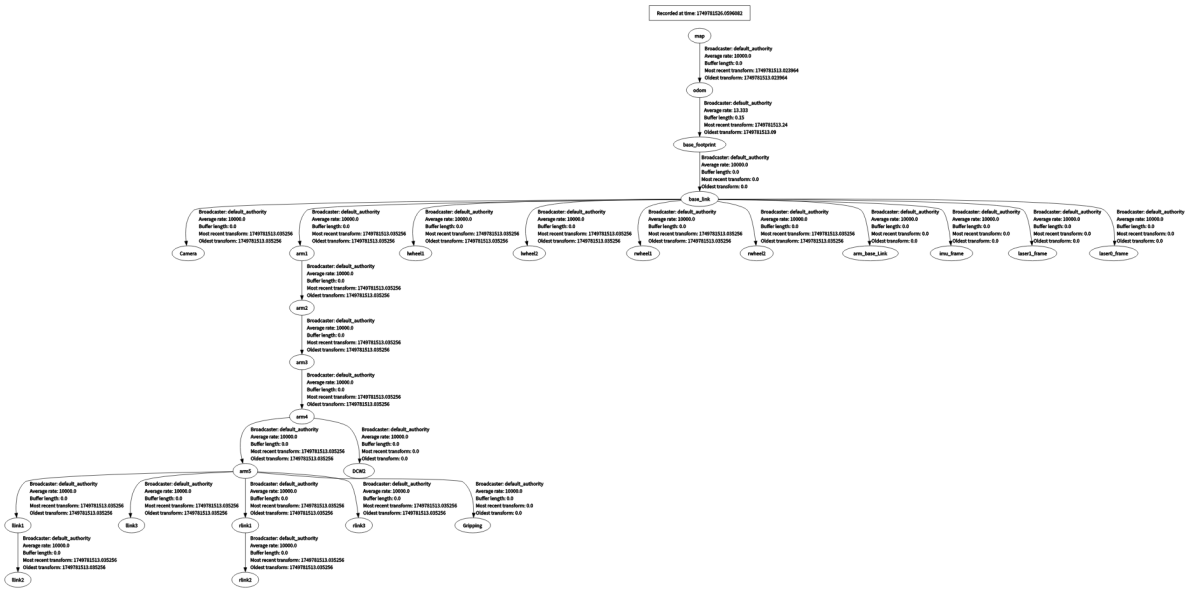



## 5.2 TF Transformation

Run in the virtual machine terminal:

```
ros2 run rqt_tf_tree rqt_tf_tree
```

The image size is too large. The original image can be viewed in this lesson's course folder.




## 5.3 GMapping Node Details

```
ros2 node info /slam_gmapping
```

Entering the above command in the terminal allows you to view the topics subscribed to and published by the GMapping node.

```
/slam_gmapping
  Subscribers:
    /parameter_events: rcl_interfaces/msg/ParameterEvent
```

```
    /scan: sensor_msgs/msg/LaserScan
  Publishers:
    /entropy: std_msgs/msg/Float64
    /map: nav_msgs/msg/OccupancyGrid
    /map_metadata: nav_msgs/msg/MapMetaData
    /parameter_events: rcl_interfaces/msg/ParameterEvent
    /rosout: rcl_interfaces/msg/Log
    /tf: tf2_msgs/msg/TFMessage
  Service Servers:
    /slam_gmapping/describe_parameters: rcl_interfaces/srv/DescribeParameters
    /slam_gmapping/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
    /slam_gmapping/get_parameters: rcl_interfaces/srv/GetParameters
    /slam_gmapping/list_parameters: rcl_interfaces/srv/ListParameters
    /slam_gmapping/set_parameters: rcl_interfaces/srv/SetParameters
    /slam_gmapping/set_parameters_atomically:
rcl_interfaces/srv/SetParametersAtomically
  Service Clients:

  Action Servers:

  Action Clients:
```

## 5.4 Configuration File

- Configuration file path:

```
~/yahboomM3_ws/user_ws/src/m3_bringup/config/slam_gmapping.yaml
```

- Default configuration parameters:

```yaml
slam_gmapping:
  ros__parameters:
    # Angular update threshold (radians): Update the map when the robot rotates
more than this angle
    angularUpdate: 0.25
    # Angular sampling step (radians): Angular search step for scan matching
    astep: 0.05
    # Robot base coordinate frame name
    base_frame: base_footprint
    # Map coordinate frame name
    map_frame: map
    # Odometry coordinate frame name
    odom_frame: odom
    # Iteration termination condition for scan matching: Stop iterating when
parameter change is less than this value
    delta: 0.05
    # Maximum number of iterations for scan matching
    iterations: 5
    # Kernel size for scan matching, affects matching smoothness
    kernelSize: 1

    # Distance sampling range for large-scale scan matching (meters)
    lasamplerange: 0.005
    # Distance sampling step for large-scale scan matching (meters)
```

```
    lasamplestep: 0.005
    # Linear update threshold (meters): Update the map when the robot moves more
than this distance
    linearUpdate: 0.5
    # Distance sampling range for small-scale scan matching (meters)
    llsamplerange: 0.01
    # Distance sampling step for small-scale scan matching (meters)
    llsamplestep: 0.01

    # Measurement noise covariance for laser scan
    lsigma: 0.075
    # Number of laser beams to skip: 0 means use all laser points
    lskip: 0
    # Linear sampling step (meters): Linear search step for scan matching
    lstep: 0.05
    # Map update time interval (seconds)
    map_update_interval: 3.0
    # Maximum laser range (meters): Points beyond this distance are ignored
    maxRange: 6.0
    # Maximum usable laser range (meters): Maximum distance used for mapping
    maxUrange: 4.0
    # Minimum score threshold for scan matching: Matches below this score are
discarded
    minimum_score: 0.0
    # Occupancy probability threshold: Cells with a probability greater than this
are considered occupied
    occ_thresh: 0.25
    # Gain factor: Affects the speed of map updates
    ogain: 3.0
    # Number of particles for the particle filter
    particles: 30
    # QoS (Quality of Service) parameter settings
    qos_overrides:
      /parameter_events:
        publisher:
          depth: 1000          # Message queue depth
          durability: volatile # Durability policy
          history: keep_all    # History policy
          reliability: reliable # Reliability policy
      /tf:
        publisher:
          depth: 1000
          durability: volatile
          history: keep_last   # Keep only the latest message
          reliability: reliable
    # Resampling threshold: Trigger resampling when the number of effective
particles is below this ratio
    resampleThreshold: 0.5
    # Angular noise covariance for scan matching
    sigma: 0.05
    # Odometry model parameter: Proportional linear error from linear motion
    srr: 0.1
    # Odometry model parameter: Proportional angular error from linear motion
    srt: 0.2
    # Odometry model parameter: Proportional linear error from angular motion
    str: 0.1
    # Odometry model parameter: Proportional angular error from angular motion
    stt: 0.2
```

```
    # Time update interval (seconds): Update the map even if the robot has not
moved
    temporalUpdate: 1.0
    # Coordinate transform publication period (seconds)
    transform_publish_period: 0.05
    # Whether to use simulation time
    use_sim_time: false
    # Maximum x-range of the map (meters)
    xmax: 100.0
    # Minimum x-range of the map (meters)
    xmin: -100.0
    # Maximum y-range of the map (meters)
    ymax: 100.0
    # Minimum y-range of the map (meters)
    ymin: -100.0
```

These are the configurable parameters for GMapping. If you need to modify the configuration parameters, you must recompile the `slam_gmapping` package in the `M3Pro_ws` workspace for the changes to take effect:

```
colcon build --packages-select slam_gmapping
```